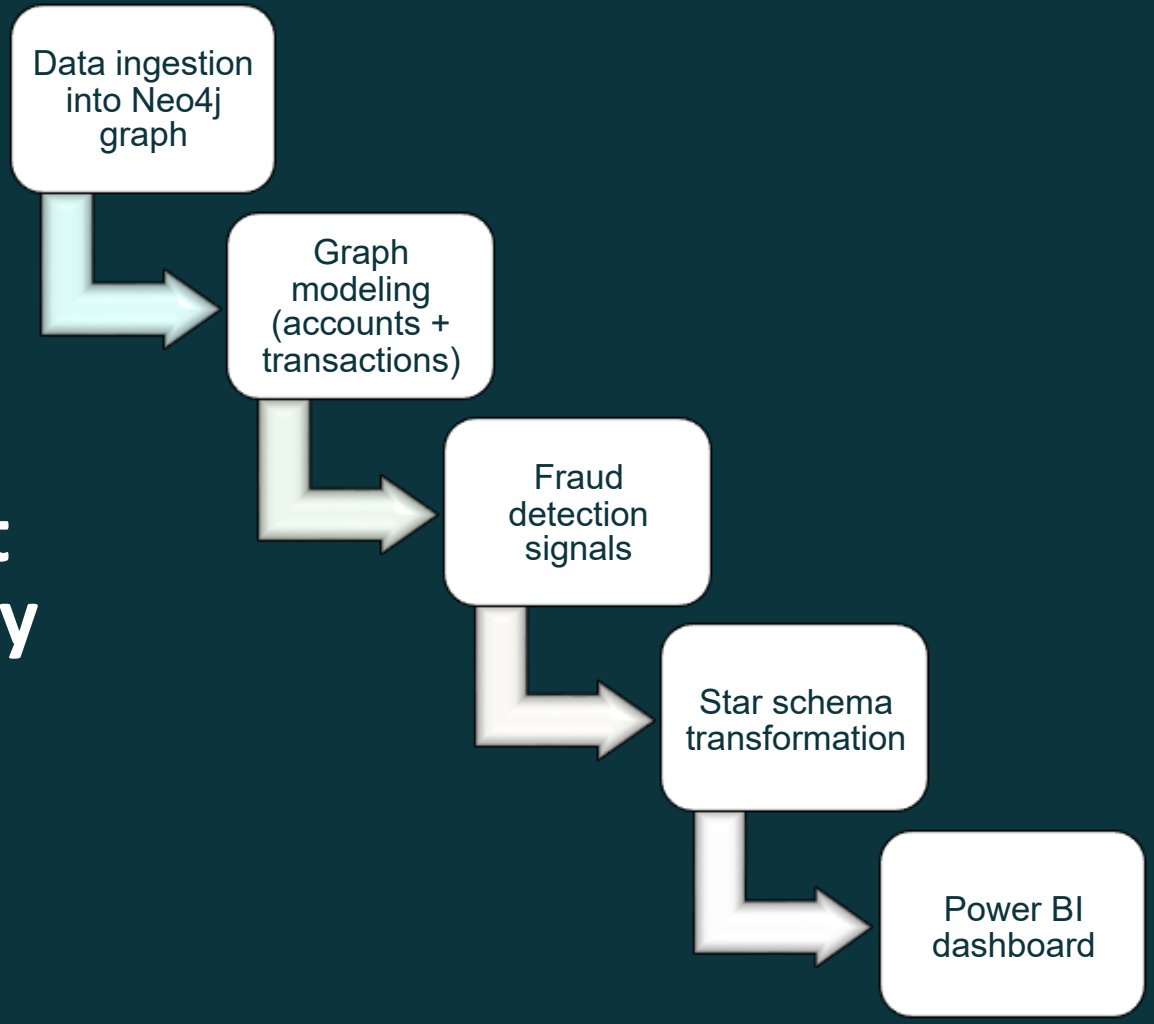
P1: Graph based Fraud Detection

By: Akanksha Satya & Jonathan Urena Salazar

Project Journey



Problem & Motivation

Why Fraud Detection Matters?

Fraud is rarely isolated behavior

Often involves networks of accounts

Traditional row-based analysis misses relationships

Detection Signal Overview

Transfer → Cash-Out

Rapid Movement Pattern

- Transfer followed by cash-out within 1–2 steps
- Strong indicator of fraud behavior

Fan-In / Fan-Out

Unusual Connectivity

- Accounts with abnormal number of connections
- Threshold: > 3 standard deviations above mean
- Grouped by time step for short-term spikes

Drain Behavior

Rapid Outflow

- Account receives funds
- Quickly transfers them out
- Time window: 10 steps

Dense Communities (Rings)

Coordinated Groups

- Louvain clustering used to detect communities
- Small groups (3–15 accounts)
- High internal transaction volume
- High internal/external ratio

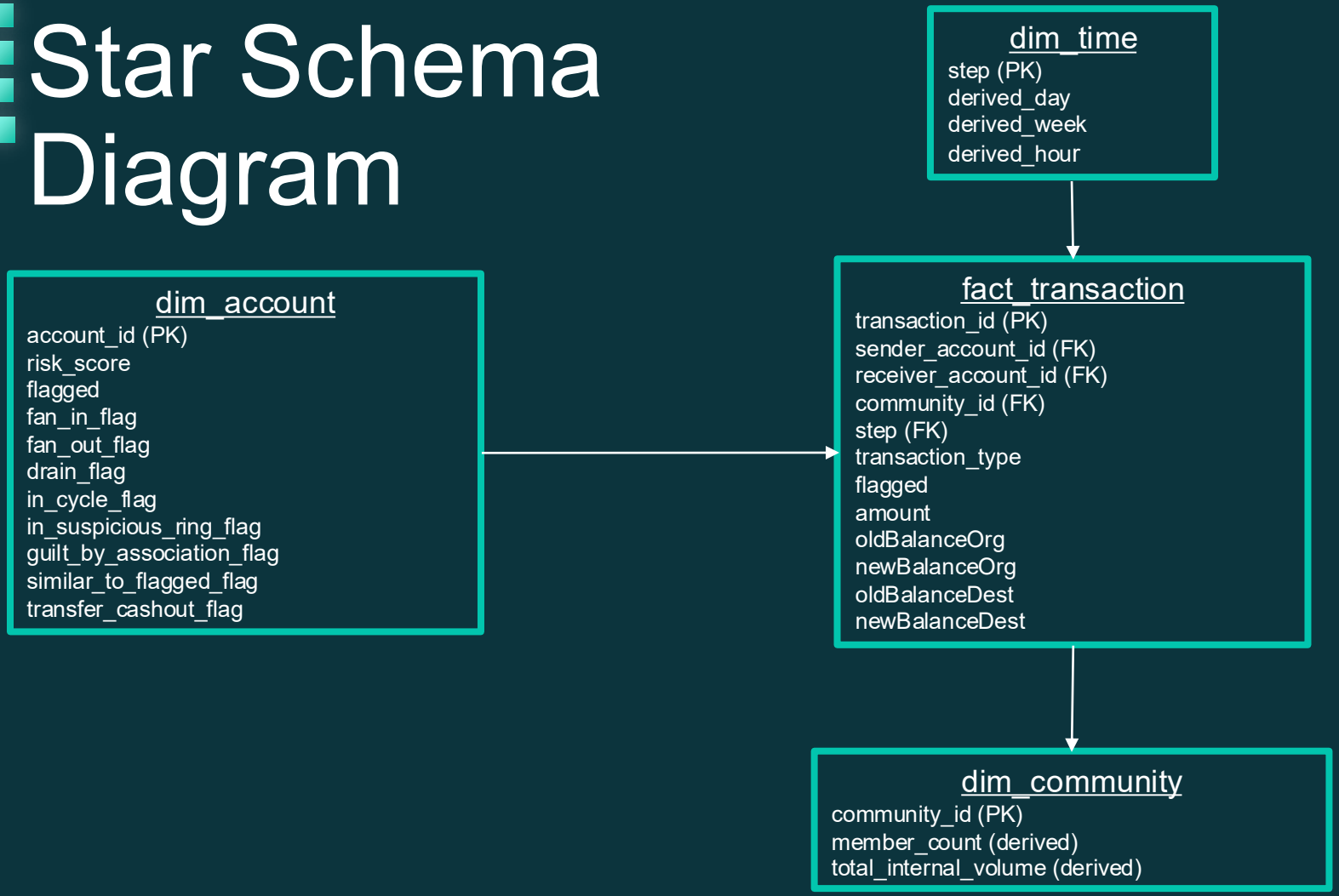
Guilt by Association

Network-Based Detection

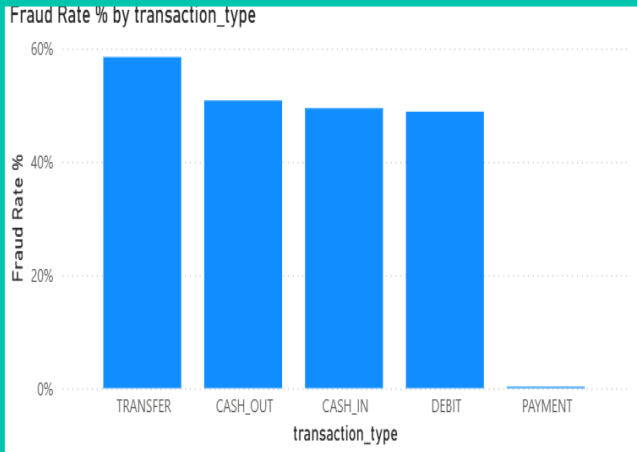
- Accounts flagged if connected to flagged accounts
- Threshold: ≥ 1 flagged neighbor

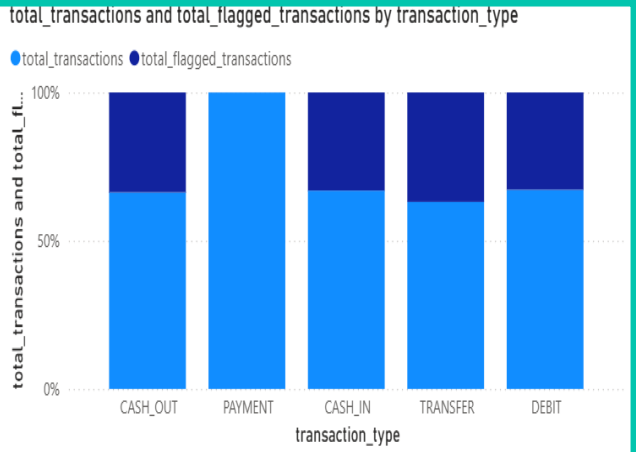
A risk score was calculated per each signals weight contribution.

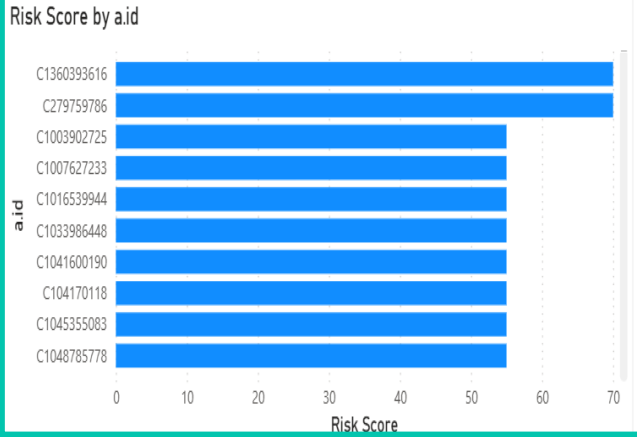
Star Schema Diagram

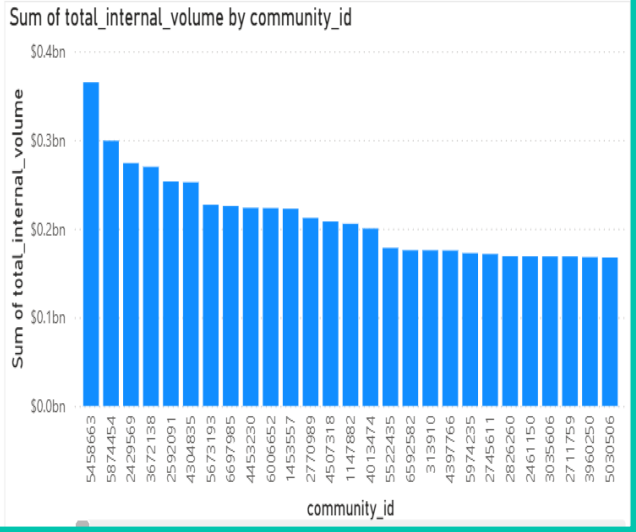


Power BI Dashboard Walkthrough

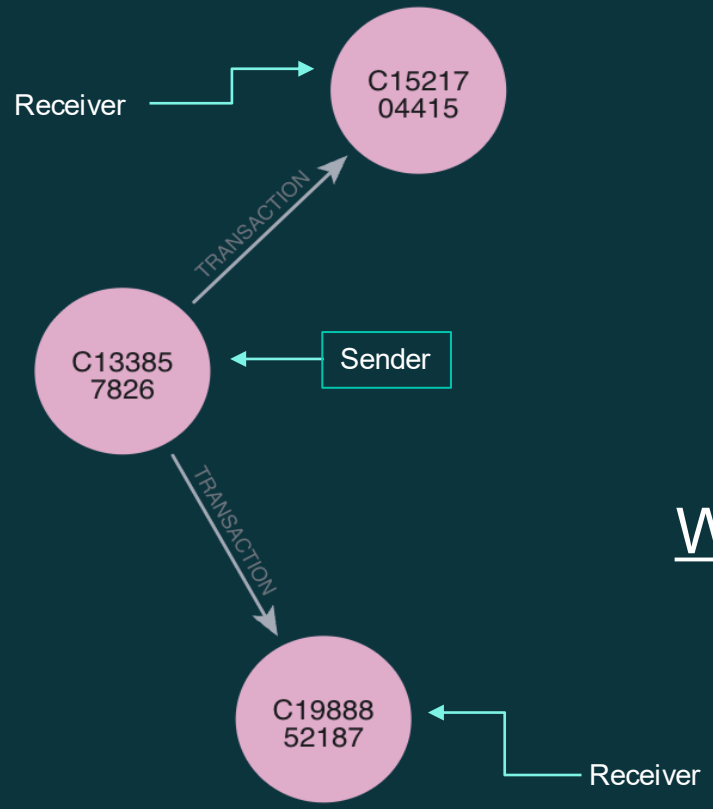








Case Study #1: Fan-Out Behavior

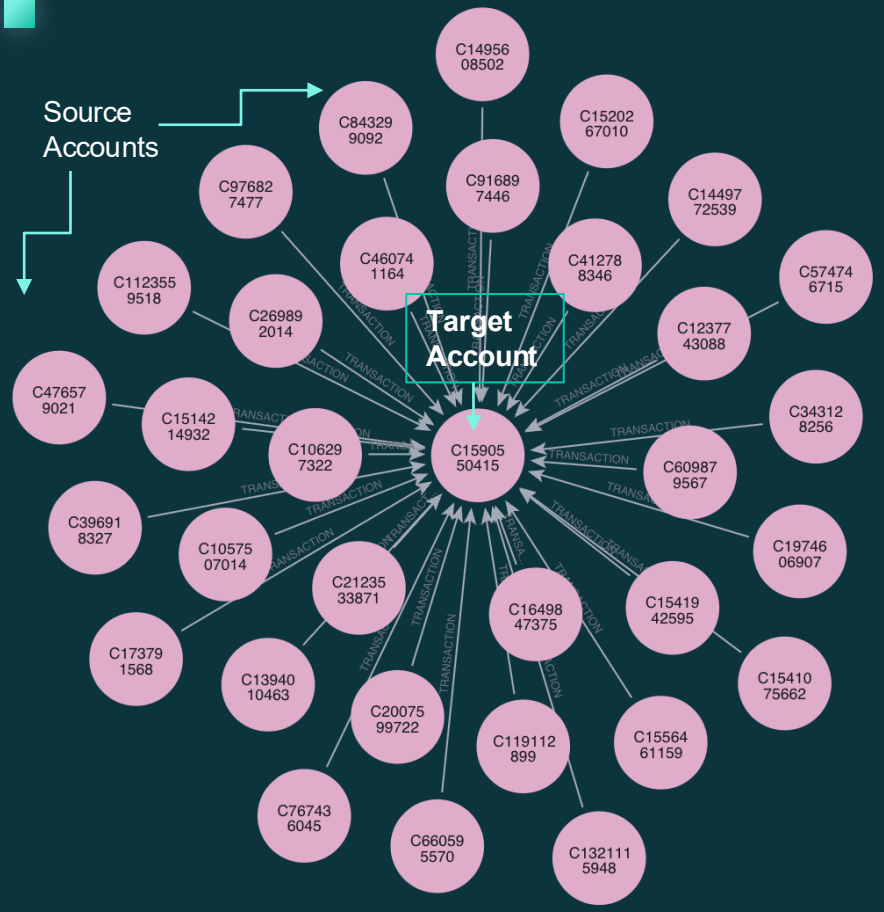


- Account: **C133857826**
- Step: **17**
- Sent to **2 receivers in same time step**
- Pattern: **Fan-Out**
- Suggests **fund distribution behavior**

Why it matters?

- Unusual compared to normal one-to-one transactions

Case Study #2: Fan-In Behavior

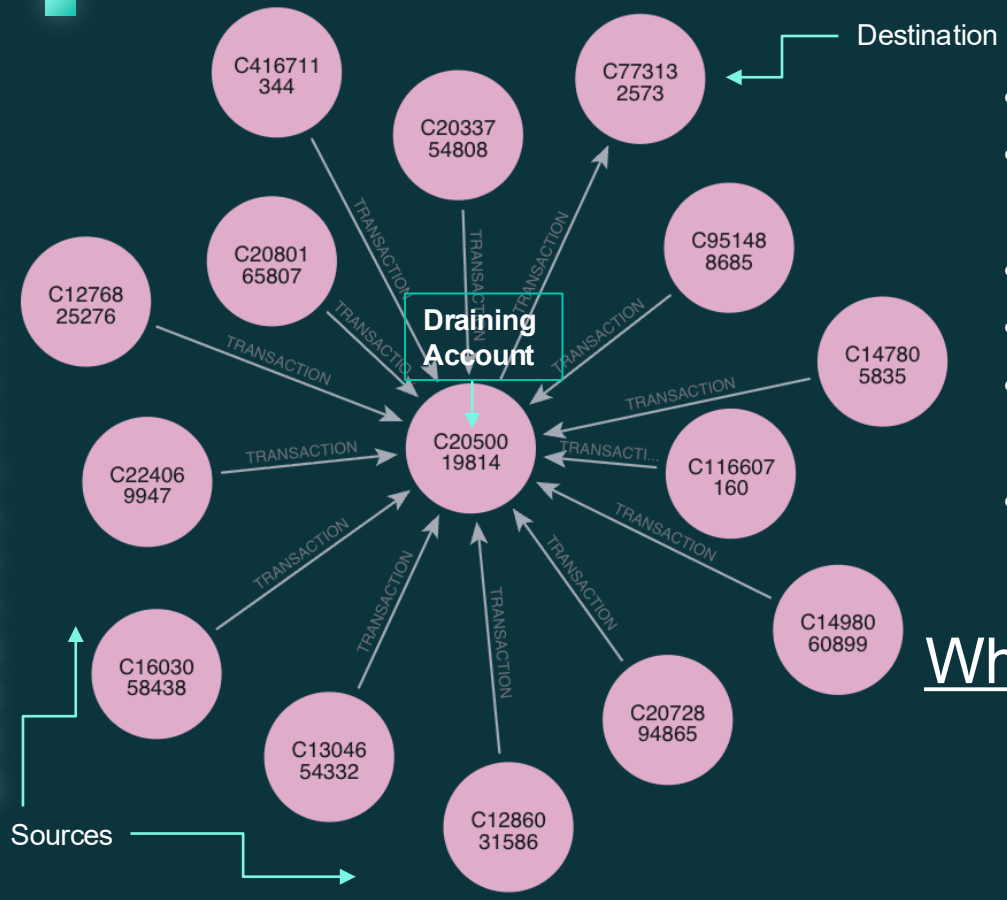


- Account: **C1590550415**
- Step: **1**
- Received from **32 unique accounts**
- Pattern: **Fan-In (Aggregation)**
- Indicates **centralized fund collection**

Why it matters?

- Unusually high number of incoming transactions in a short time window

Case Study #3: Drain Behavior



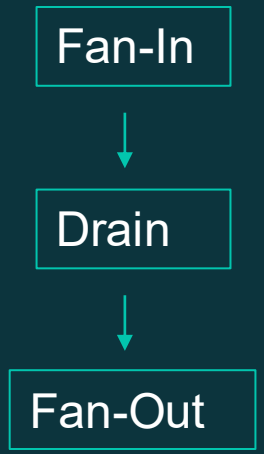
- Account: **C2050019814**
- Receives **funds from multiple sources**
- Quickly transfers funds out
- Time window: **≤ 10 steps**
- Leaves very **low remaining balance**
- Pattern: **Drain** (Pass-through behavior)

Why it matters?

- Acts as an intermediary rather than a final destination

Case Study Summary

- **Fan-Out (Distribution)**
  - One account sends funds to multiple recipients
  - Suggests **fund splitting / layering**
- **Fan-In (Aggregation)**
  - Multiple accounts send funds to one account
  - Indicates **centralized collection (hub)**
- **Drain Behavior (Intermediary)**
  - Account receives and quickly transfers funds
  - Suggests **pass-through / laundering activity**



Fraud is not a single action; it is a coordinated flow of funds across multiple accounts.

What Worked

- Graph model effectively captured relationships
- Transfer → cash-out signal was strong
- Community detection revealed clusters
- Efficient data processing with Pandas + PyArrow (Parquet)
- Pipeline to Power BI worked end-to-end

What Didn’t Worked

- Node similarity (Jaccard) not effective
- Cycles not present in dataset
- Some signals didn’t produce results

Challenges

Memory constraints (Louvain, similarity)

- Adjusted Neo4j memory configuration (RAM limits)

Had to batch updates in Neo4j

Precomputed statistics for fan-out

Thresholds too high → over-flagging of accounts

Lessons Learned

Graph analysis > traditional methods for fraud

Real datasets don’t match theory perfectly

Threshold tuning is critical

Data modeling matters for visualization

Thanks!

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